



THE UNIVERSITY OF CHICAGO

REWARD AND GOAL COMPETITION IN SUSTAINED
ATTENTION: EVIDENCE FROM EEG AND
MULTIVARIATE DECODING

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Abstract

Sustained attention enables goal-directed behavior over extended periods but is frequently disrupted by attentional lapses. Recent accounts propose that such lapses arise from competition between task goals rather than simple failures of cognitive control. The present study examined how reward and task switching interact to shape sustained attention and its neural mechanisms. Thirty participants completed a rewarded bilateral switch Continuous Performance Task (switch-CPT) while electroencephalography (EEG) was recorded. Behaviorally, switching between task goals increased no-go errors, particularly when switching to an unrewarded goal, and these switch costs were strongest early in each block. Reward selectively reduced these errors, especially under conditions of high goal competition.

Neural results revealed a dissociation across processing stages. Early spatial attention, indexed by the N2pc, was sensitive to trial type but not reliably modulated by reward or switching. In contrast, later control-related processing, indexed by the P3, was strongly influenced by both factors, with larger amplitudes for no-go trials, reduced responses under switching, and enhanced responses under reward. Oscillatory analyses showed that frontocentral theta power was greater for no-go errors than correct trials, indicating increased control demands at the time of failure, whereas alpha activity did not track attentional lapses. Multivariate pattern analysis further demonstrated that reward context could be decoded from distributed EEG activity both within and across participants.

Together, these findings support a dynamic account of sustained attention in which attentional lapses emerge from goal competition and are most strongly reflected in later-stage control processes rather than early perceptual selection. Reward enhances performance by stabilizing goal engagement, particularly when competition between goals is high.

Keywords: Sustained attention; reward; EEG; goal competition; MVPA

1 Introduction

Sustained attention is the ability to maintain goal-directed behavior over time, even when tasks are repetitive or distractions are present. Despite its importance for learning, work, and safety, people frequently experience brief lapses in which behavior drifts from the intended goal. One recent account proposes that these lapses arise from changes in opportunity cost—the brain’s continuous assessment of the value of staying on task relative to alternative actions (Esterman et al., 2014). When the opportunity cost of persisting with the current task increases, engagement declines and the likelihood of distraction rises.

Building on this idea, Chidharom, Vogel, and Rosenberg (2025) proposed a goal-competition hypothesis suggesting that lapses occur when goals compete. In this view, multiple goals are stored in long-term memory and compete for access to working memory based on their perceived reward value. According to this framework, the brain continuously compares the expected value of available goals and allocates attention to the one that appears most rewarding. A lapse occurs when an alternative goal temporarily wins this competition over the current task goal, meaning that distraction is not merely a failure of control but a shift in priority toward a competing goal (Chidharom, Vogel, & Rosenberg, 2025).

To test this hypothesis, the authors employed the switch Continuous Performance Task (switch-CPT). In this paradigm, participants alternate between two task goals (face vs. scene) across blocks. On each trial, a face and a scene are presented simultaneously, and participants respond to one subcategory within the cued goal (e.g., male faces) while withholding responses to the other (e.g., female faces). Because task switching requires participants to disengage from one goal and activate another, it increases competition between active goals, making this paradigm ideally suited to test whether goal competition alone can produce lapses of sustained attention. Behaviorally, participants made more errors on no-go trials (i.e., failures to withhold a response when required) when they had to switch to a new goal than when they repeated the same one. These no-go errors mainly occurred during the first few trials after the switch cue and then gradually declined. This pattern shows that switching goals temporarily increases competition between the old and new goals. While the previous goal is still active in memory, it interferes with the new one, making it harder to stay focused. In repeat blocks, where the same goal continues, there is little competition and performance remains more stable (Chidharom, Vogel, & Rosenberg, 2025).

In another version, Chidharom, Vogel, and Rosenberg (2025) tested the effect of opportunity cost on the occurrence of lapses. To do so, they monetarily rewarded one goal (e.g., attending to faces) but not the other (e.g., attending to scenes) during the same switch-CPT. The results showed that participants made fewer errors on no-go trials when performing the rewarded goal compared to the unrewarded goal, indicating that motivation reduced lapses of sustained attention. Crucially, the typical switch cost, an increase in errors immediately after a goal change, was observed only when participants switched from the rewarded to the unrewarded goal, and not when switching to the rewarded one. This suggests that reward lowers the opportunity cost of maintaining the current task and reduces competition from alternative goals (Chidharom, Vogel, & Rosenberg, 2025). Together, these findings suggest that reward value influences goal competition and results in distraction, but the neural mechanisms underlying this process remain unknown. This leads to the central question of the present proposal: through which cognitive and neural mechanisms does switching toward an unrewarded goal (high opportunity cost) increase lapses.

To investigate how reward and task switching influence sustained attention, we used a bilateral switch Continuous Performance Task (switch-CPT). On each trial, a letter and a digit were presented simultaneously on either side of fixation, and participants responded based on a cued task rule (letter or number) that changed across blocks. This design allows precise control over perceptual input while isolating goal selection and attentional processes under varying reward conditions. One task rule was associated with monetary reward. In rewarded blocks, correct no-go responses earned a small bonus (+0.04), whereas missed responses on go trials were penalized. We use EEG to track rapid changes to better characterize neural mechanisms during the goal switch in the switch CPT task. Here two event-related potentials (ERPs) index the cognitive process of interest: the parietal P3, a well-established marker for attentional allocation and engagement and the N2pc, a spatial neural marker for spatial attention to goal-relevant locations (Donchin & Coles, 1988; Kam et al., 2022). The P3 is quantified as the mean amplitude at parietal electrode Pz within a 300–700 ms window following stimulus onset. The N2pc is computed as the difference in activity between contralateral and ipsilateral posterior electrodes (P7/P8) in the 200–300 ms time window, relative to the onset of the task-relevant stimulus.

In addition to ERP measures, we examined oscillatory dynamics to capture fluctuations in attentional state. Prior work has shown that changes in EEG oscillatory activity are associated with attentional lapses, with increases in posterior alpha linked to disengagement and frontocentral theta associated with conflict monitoring and control processes (O’Connell et al., 2009). Based on this framework, we expected that theta activity during stimulus processing would be greater for no-go error trials, reflecting increased control demands, whereas alpha activity would reflect fluctuations in attentional engagement.

Under the theory of goal competition, we can make specific predictions about the rewards and switching. On the behavioral level, when participants switch away from a previously rewarded goal (high opportunity cost), the alternative rewarded goal still remains important. This creates a goal competition and therefore increases the likelihood of errors on no-go trials. Under these conditions, we expected reduced P3 from lower engagement with the newly cued goal. When participants switch toward a rewarded goal from an unrewarded goal (low opportunity cost), we expect larger P3 and stronger N2pc to goal-relevant locations, reflecting better engagement and selection.

Beyond ERP analyses, we conducted multivariate pattern analysis (MVPA) to examine whether reward condition (rewarded vs. unrewarded) could be decoded from distributed EEG activity. Specifically, a classifier was trained to distinguish between rewarded and unrewarded trials based on trial-level neural patterns. This approach tests whether reward-related information is reflected in the spatial distribution of EEG activity during task performance. If reward modulates attentional processing, neural patterns should differ system-

atically across reward conditions. Together, these analyses complement ERP measures by capturing distributed neural signals associated with reward and task performance.

2 Methods

2.1 Participants

Thirty healthy adults (ages 18–35) were recruited from the University of Chicago community through university postings and SONA participant pools. All participants had normal or corrected-to-normal vision and reported no history of neurological disorders (e.g., epilepsy, stroke) or psychiatric conditions. Each was provided informed consent prior to participation. Compensation included hourly pay plus task-contingent rewards, for a total of approximately 60–70 per session. All procedures had IRB approval from the University of Chicago review board. One participant ($M = 0.61$) was identified as an outlier based on a criterion of ± 2.5 standard deviations from the group mean no-go error rate ($M = 0.221$, $SD = 0.117$) and was excluded from further analyses.

2.2 Stimuli

Stimuli consisted of images drawn from four categories corresponding to the task conditions (even, odd, vowel, consonant). On each trial, two images were presented simultaneously to the left and right of a central fixation cross, with one image serving as the task-relevant stimulus and the other as a distractor. Each image was displayed at a size of 300×300 pixels and positioned approximately 200 pixels to the left or right of fixation. The spatial position of the task-relevant stimulus was randomized across trials. Task cues were presented as colored squares (100×100 pixels) at fixation at the beginning of each mini-block. Cue color (red or blue) indicated the currently relevant task, with cue–task mappings counterbalanced across participants.

Go and no-go conditions were defined based on the active task rule. In each task, one category required a response (go) and the other required response withholding (no-go), with no-go trials occurring on 20% of trials.

2.3 Procedure

Participants completed a rewarded bilateral switch Continuous Performance Task (switch-CPT) that alternates between two goal conditions: a number task (even/odd) and a letter task (vowel/consonant). At the start of each block, a colored cue (red or blue) appears to indicate which task the participant performs.; either a number or a letter block. (Figure 1)

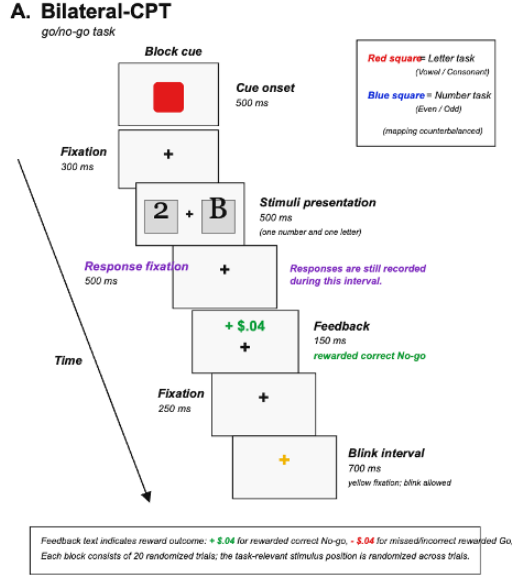


Figure 1: Task and behavioral performance. Bilateral switch CPT. Participants were instructed to focus on either letters or numbers depending on a cue presented every 20 trials.

Each block consisted of 20 trials. Trials began with a fixation display (300 ms), followed by stimulus presentation, during which one letter and one number were presented simultaneously for 500 ms. After stimulus offset, a fixation display remained on screen for an additional 500 ms, during which responses were still recorded. Participants then received feedback for 150 ms, followed by a 250-ms fixation display and a 700-ms blink interval. Trial order within each block was randomized, and the spatial position of the relevant stimulus was randomized across trials. Prior to the main experiment, participants completed a structured training session that included practice blocks for each task, followed by practice with both repeat and switch trials. Participants were required to reach a minimum accuracy threshold before proceeding to the main task. The main task consisted of 128 blocks, yielding a total of 2,560 trials. Blocks were arranged in a balanced sequence including both task repetitions and task switches. Reward was manipulated at the block level. One task (counterbalanced across participants) was designated as rewarded, such that correct no-go responses yielded a small monetary bonus (0.04), while missed responses on go trials incurred an equivalent penalty. The other task was unrewarded and provided feedback without monetary consequences. The experiment lasted approximately three hours, including EEG setup, instructions, practice, and breaks.

2.4 EEG Recording

Continuous EEG activity was recorded in a sound-attenuated chamber using 30 scalp electrodes mounted in an elastic cap with two additional mastoid electrodes (actiCHamp, Brain Products, Munich, Germany). Electrodes were positioned according to the international 10–20 system at the following sites: Fp1, Fp2, F7, F3, Fz, F4, F8, FC5, FC1, FC2, FC6, C3, Cz, C4, CP5, CP1, CP2, CP6, P7, P3, Pz, P4, P8, PO7, PO3, PO4, PO8, O1, Oz, and O2. Two additional electrodes were placed on the left and right mastoids, and the ground electrode was positioned at Fpz. (Figure 2) Signals were referenced online to the right mastoid and re-referenced offline to the algebraic average of the left and right mastoids. Electrooculogram (EOG) activity was recorded using passive electrodes. Horizontal EOG was measured using a bipolar configuration with electrodes placed approximately 1 cm from the outer canthus of each eye. Vertical EOG was recorded using a bipolar pair placed above and below the right eye. Data were filtered online with a bandpass of 0.01–80 Hz (12 dB/octave slope) and digitized at a sampling rate of 1000 Hz using BrainVision Recorder (Brain Products, Munich, Germany). Gaze position was recorded using a desk-mounted infrared eye-tracking system (EyeLink 1000 Plus, SR Research, Ottawa, Ontario, Canada) at a sampling rate of 1,000 Hz. Participants maintained a stable head position using a chin rest. The eye tracker was calibrated at the beginning of the experiment and recalibrated after each break.

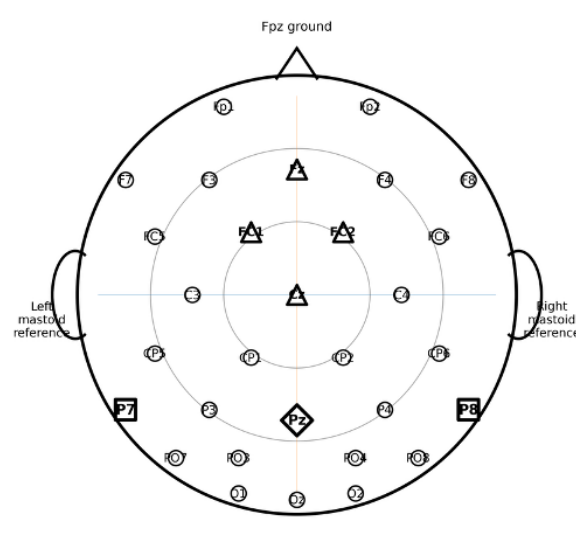


Figure 2: Scalp electrode locations based on the international 10–20 system. Key electrodes used in analyses are highlighted: posterior lateral electrodes (P7/P8) for N2pc, and the parietal electrode (Pz) for P3. Mastoid electrodes served as reference, and Fpz was used as ground.

2.5 EEG Processing

EEG data were preprocessed using EEGLAB and ERPLAB. Eye-tracking data were synchronized with EEG recordings to enable rejection of trials containing gaze deviations or saccades. Data were epoched from 200 to 950 ms. Artifact rejection was performed using an automated detection procedure followed by manual visual inspection. Trials containing ocular artifacts (saccades or blinks), excessive voltage fluctuations (± 100 V), step-like changes, linear drifts, high-frequency muscle noise, or flatlined channels were excluded from EEG analyses. Trials were excluded if they contained ocular artifacts (blinks or saccades), excessive voltage fluctuations exceeding ± 100 V, abrupt step-like changes in voltage, or abnormal trends such as linear drifts. In addition, trials contaminated by high-frequency muscle noise or containing flatlined channels were removed. Artifact detection was implemented using moving-window peak-to-peak thresholds and step-function criteria to identify transient noise, ensuring that only clean trials were included in ERP analyses. On average, about 26% of trials were rejected for each participant.

For ERP analyses, the N2pc was quantified as the difference in mean amplitude between contralateral and ipsilateral electrodes relative to the location of task-relevant stimuli, focusing on P7 and P8 within the 200–400 ms post-stimulus window. The P3b was measured at electrode Pz within the 300–700 ms post-stimulus window. Mean amplitudes were extracted within ± 50 ms windows centered on the component peak latency. Both peak amplitude and latency were extracted for each participant and condition.

2.6 Data Analysis

Behavioral analysis Behavioral performance was primarily assessed using no-go error rates, defined as failures to withhold a response on no-go trials. This measure was used because no-go errors provide a direct index of attentional lapses, reflecting moments in which the currently relevant task goal fails to guide behavior under conditions of goal competition. To examine the effects of task switching and reward, no-go error rates were analyzed using a repeated-measures 2×2 analysis of variance (ANOVA) with Condition (switch vs. repeat) and Reward (rewarded vs. unrewarded) as within-subject factors. The critical test of interest was the Condition $\times$ Reward interaction, which evaluates whether attentional lapses increase selectively when participants switch toward an unrewarded goal compared to switching toward a rewarded goal. To further characterize this interaction, planned paired-samples t-tests were conducted to compare switch and repeat conditions separately within each reward context. In particular, we tested whether switch costs (switch minus repeat error rates) were greater in unrewarded blocks than in rewarded blocks. These comparisons provide a more direct test of the hypothesized asymmetry in goal competition

across reward conditions.

While the ANOVA provides a condition-level summary of behavioral performance, it relies on aggregated data and does not fully capture trial-by-trial variability or the binary nature of accuracy outcomes. In addition to these aggregated analyses, trial-level behavior was analyzed using generalized linear mixed-effects models (GLMMs) with a binomial distribution and logit link function. In the primary model, trial-level accuracy (correct vs. error) was predicted from Trial Type (go vs. no-go), Reward Block (rewarded vs. unrewarded), and Switch Block (switch vs. repeat), including all interactions among these factors, with a random intercept for subject. A second GLMM focused specifically on no-go trials to examine within-block dynamics of attentional control. In this model, accuracy on no-go trials (correct vs. error) was predicted from centered trial number within block, Reward Block, and Switch Block, including all interactions, with a random intercept for subject. This analysis allowed us to assess not only overall effects of reward and switching, but also how these effects evolved over time within a block, providing a more fine-grained characterization of goal competition dynamics.

EEG Analysis: N2pc To examine whether lateralized attentional selection varied as a function of trial outcome, task switching, and reward, N2pc amplitudes were analyzed using a $2 \times 2 \times 2$ repeated-measures design with within-subject factors of Trial Type (go, no-go), Condition (repeat, switch), and Reward (rewarded, unrewarded). The N2pc was measured at posterior electrodes P7/P8 and was defined as contralateral minus ipsilateral activity relative to the stimulus side. For each participant, contralateral and ipsilateral waveforms were derived based on stimulus laterality, and N2pc waveforms were then computed from these lateralized signals. For each Trial Type $\times$ Condition $\times$ Reward cell, the negative N2pc peak was identified within a 200–400 ms post-stimulus search window. Mean amplitude was then extracted within a ± 50 ms window centered on the individual peak latency. This yielded one N2pc amplitude value per participant for each of the 12 within-subject cells. These values were entered into a repeated-measures ANOVA testing the main effects of Trial Type, Condition, and Reward, as well as all two-way and three-way interactions. Follow-up pairwise comparisons were conducted to further examine significant omnibus effects.

To further decompose the effects observed in the N2pc analysis, follow-up analyses were conducted separately on contralateral (contra) and ipsilateral (ipsi) activity. These analyses aimed to determine whether modulation of the N2pc was driven primarily by changes in contralateral enhancement, ipsilateral suppression, or both. For each participant, contra and ipsi waveforms were derived from the same posterior electrode pairs (P7/P8), based on stimulus laterality. Mean amplitudes were extracted using the same procedure as the N2pc analysis: a 200–400 ms search window was used to identify the negative peak, and mean

amplitude was computed within a ± 50 ms window around the individual peak. Separate $2 \times 2 \times 2$ repeated-measures ANOVAs were conducted on contra and ipsi amplitudes, with within-subject factors of Trial Type (go correct, no-go correct, no-go error), Condition (repeat, switch), and Reward (rewarded, unrewarded).

EEG Analysis: P3 To examine how attentional engagement and control processes varied as a function of trial outcome, task switching, and reward, P3 amplitudes were analyzed using a $2 \times 2 \times 2$ repeated-measures design with within-subject factors of Trial Type (go, no-go), Condition (repeat, switch), and Reward (rewarded, unrewarded). The P3 was measured at the midline parietal electrode (Pz), where it is typically maximal.

For each participant, mean P3 amplitude was extracted within a 300–700 ms post-stimulus window. Consistent with the N2pc procedure, peak latency was first identified within this window, and mean amplitude was then computed within a ± 50 ms window centered on the individual peak. This yielded one P3 amplitude value per participant for each condition cell. These values were entered into a repeated-measures ANOVA testing the main effects of Trial Type, Condition, and Reward, as well as all two-way and three-way interactions. Follow-up pairwise comparisons were conducted to further examine significant omnibus effects

Oscillatory Power Analysis To examine oscillatory dynamics associated with attentional lapses, band-limited power analyses were conducted using a bandpass filtering and Hilbert transform approach. This analysis was motivated by prior work demonstrating that fluctuations in EEG oscillatory activity can predict attentional failures well before behavioral errors occur. In particular, increases in posterior alpha-band activity have been shown to precede missed targets by several seconds to tens of seconds, reflecting gradual disengagement of attentional control processes (O’Connell et al., 2009). In contrast, theta-band activity has been linked to conflict monitoring and cognitive control processes that emerge at the time of stimulus processing (O’Connell et al., 2009). Based on this framework, we hypothesized that: frontocentral theta power during stimulus processing would be greater for no-go error trials than correct trials, reflecting increased conflict or failed inhibitory control, and posterior alpha power would index fluctuations in attentional engagement, with potential differences emerging prior to or during stimulus processing.

EEG data were epoched from 200 to 950 ms relative to stimulus onset, consistent with the epoch used for ERP analyses. Posterior alpha-band activity (8–14 Hz) was computed by averaging signals across parietal–occipital electrodes (Pz, P3, P4, PO3, PO4, PO7, PO8, O1, Oz, O2), while frontocentral theta-band activity (4–7 Hz) was computed from electrodes Fz, FC1, FC2, and Cz. For each trial, signals were bandpass filtered using a fourth-order

Butterworth filter and transformed using the Hilbert transform to obtain instantaneous power (squared analytic amplitude). Mean power was extracted for a pre-stimulus window (200 to 0 ms) and a post-stimulus window (0 to 950 ms).

Analyses focused on no-go trials, comparing correct inhibitions and commission errors. For each participant, power values were averaged within each condition defined by accuracy (error vs. correct), condition (switch vs. repeat), and reward (rewarded vs. unrewarded), yielding one value per cell. These subject-level estimates were analyzed using linear mixed-effects models with accuracy, condition, and reward as fixed effects and subject as a random intercept

MVPA Decoding To examine whether reward-related information could be decoded from EEG activity, a multivariate pattern analysis (MVPA) was conducted on trial-level data. Continuous EEG signals were epoched from 200 to 500 ms relative to stimulus onset. For each trial, mean EEG amplitude within the stimulus window (0–500 ms) was extracted at each electrode (Fp1, Fp2, F3, F4, F7, F8, Fz, FC1, FC2, FC5, FC6, C3, C4, Cz, CP1, CP2, CP5, CP6, P3, P4, P7, P8, Pz, PO3, PO4, PO7, PO8, O1, O2, Oz), yielding a multivariate feature vector that captured the spatial distribution of neural activity across the scalp. Thus, each trial was represented by a single vector of electrode-wise mean activity values.

Trials were labeled according to reward condition (rewarded vs. unrewarded), and trials with missing or ambiguous labels were excluded. Classification was performed using a linear support vector machine (SVM). To assess subject-specific decoding performance, a within-subject cross-validation approach was implemented, in which trials from each participant were randomly split into training (80%) and test (20%) sets. This procedure was repeated 20 times with different random splits, and classification accuracy was averaged across repetitions for each participant.

Within each training fold, feature values were z-scored across trials, and the same normalization parameters were applied to the corresponding test data to prevent information leakage. Subject-level decoding accuracies were then compared against chance (0.5) using a one-sample t-test at the group level. One participant (Subject 8) was excluded due to excessive noise, resulting in a final sample of 29 participants.

3 Results

3.1 Behavioral performance

Participants performed the task with near-ceiling accuracy on Go trials, with an overall Go accuracy of 98.7% ($M = 0.987$, $SD = 0.011$). Go accuracy remained consistently high across

all experimental conditions, with minimal variation between repeat rewarded ($M = 0.988$, $SD = 0.013$), repeat unrewarded ($M = 0.989$, $SD = 0.010$), switch rewarded ($M = 0.987$, $SD = 0.011$), and switch unrewarded trials ($M = 0.982$, $SD = 0.016$).

Across all conditions, participants failed to withhold responses on 22.1% of no-go trials on average ($M = 0.221$, $SD = 0.117$), indicating a moderate rate of attentional lapses during the task. The lowest error rates were observed during rewarded repeat trials ($M = 0.185$, $SD = 0.108$), whereas the highest error rates occurred when switching to an unrewarded block ($M = 0.258$, $SD = 0.127$). Intermediate levels of error were observed when switching to a rewarded block ($M = 0.190$, $SD = 0.106$) and repeating unrewarded trials ($M = 0.208$, $SD = 0.109$) (Figure 3).

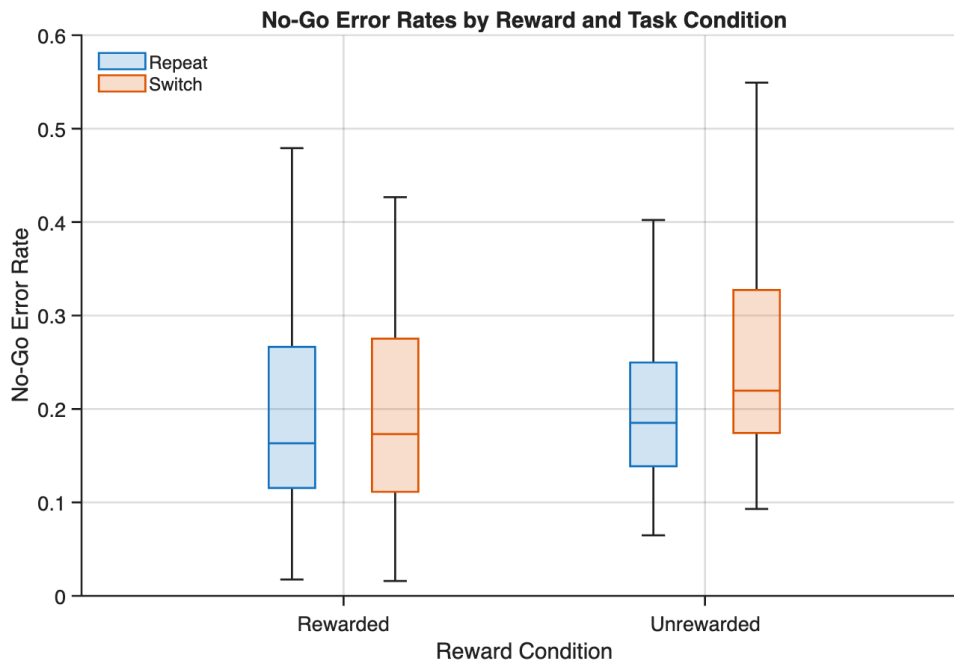


Figure 3: No-go error rates by reward and task condition.

A 2×2 repeated-measures ANOVA revealed a significant main effect of Condition, $F(1, 28) = 6.62$, $p = .016$, $\eta^2_p = .19$, indicating that participants committed more no-go errors on switch trials than on repeat trials, reflecting a switch-related cost in sustained attention. A significant main effect of Reward was also observed, $F(1, 28) = 7.12$, $p = .013$, $\eta^2_p = .20$, indicating that participants committed more no-go errors on unrewarded trials than on rewarded trials, suggesting that reward enhanced sustained attentional engagement. Importantly, the Condition $\times$ Reward interaction was significant, $F(1, 28) = 6.51$, $p = .016$, $\eta^2_p = .19$, indicating that the effect of task switching on no-go error rates differed as a function of reward.

Post-hoc analyses were conducted to examine switch costs under different reward conditions. Under unrewarded conditions, switch trials produced significantly higher no-go error rates than repeat trials ($p = .001$), indicating increased attentional lapses when participants switched task goals. In contrast, under rewarded conditions, no-go error rates did not differ significantly between switch and repeat trials ($p = .689$), suggesting that reward attenuated switch costs. Post-hoc tests further examined the effect of reward within each task condition. Reward did not significantly reduce no-go error rates during repeat trials ($p = .206$). In contrast, reward significantly reduced no-go error rates during switch trials ($p = .002$), indicating that motivational incentives particularly benefited performance under conditions of increased goal competition.

3.2 Trial-Level Mixed-Effects Analysis

To further examine these effects at the trial level, a generalized linear mixed-effects model (binomial logit link) was conducted predicting accuracy from trial type (go vs. no-go), reward (rewarded vs. unrewarded), and switching (switch vs. repeat), with random intercepts for subjects. The same outlier identified in the behavioral analysis was excluded from this model, resulting in a final sample of 29 participants. There was a robust main effect of trial type, such that NoGo trials were associated with significantly lower accuracy than Go trials, $\beta = 3.19$, $SE = 0.11$, $p < .001$. A significant main effect of switching was also observed, with reduced accuracy on switch relative to repeat trials, $\beta = 0.46$, $SE = 0.12$, $p < .001$. The main effect of reward was not significant, $\beta = 0.12$, $SE = 0.13$, $p = .369$. At the interaction level, the interaction between reward and switching was significant, $\beta = 0.43$, $SE = 0.17$, $p = .014$, indicating that the effect of reward differed as a function of switching demands. Specifically, reward-related benefits were more pronounced under switch conditions relative to repeat conditions. The interaction between trial type and reward was marginally significant, $\beta = 0.28$, $SE = 0.15$, $p = .058$, suggesting a trend toward greater reward-related improvements in accuracy on NoGo trials compared to Go trials. In contrast, the interaction between trial type and switching was not significant, $\beta = 0.21$, $SE = 0.14$, $p = .131$, indicating that the effect of switching did not differ reliably between Go and NoGo trials. Finally, the three-way interaction between trial type, reward, and switching was not significant, $\beta = 0.19$, $SE = 0.20$, $p = .325$, suggesting that the combined influence of reward and switching on performance did not differ across trial types.

3.3 Trial-Level Mixed-Effects Analysis : NoGo

To further examine how attentional lapses evolved over time and were modulated by reward and task switching, a trial-level generalized linear mixed-effects model (GLMM) was con-

ducted on NoGo trials. The analysis revealed a significant effect of trial number, $\beta = 0.020$, $SE = 0.009$, $p = .022$, indicating that NoGo accuracy decreased over time within a block, reflecting a gradual increase in attentional lapses. A significant main effect of reward was observed, $\beta = 0.147$, $SE = 0.071$, $p = .039$, indicating that accuracy was higher on rewarded trials compared to unrewarded trials. Switching also exerted a significant effect, $\beta = 0.273$, $SE = 0.068$, $p < .001$, such that accuracy was lower on switch trials relative to repeat trials, reflecting a switch-related cost, consistent with the ANOVA results.

There was no significant interaction between trial number and reward, $\beta = 0.006$, $SE = 0.012$, $p = .644$, indicating that the effect of time within a block did not differ between rewarded and unrewarded conditions. In contrast, a significant interaction between trial number and switching emerged, $\beta = 0.057$, $SE = 0.012$, $p < .001$. Follow-up examination of simple slopes indicated that NoGo accuracy declined over time in repeat trials, $\beta = 0.020$, whereas accuracy increased over time in switch trials, $\beta = 0.038$. This crossover pattern suggests that switch-related performance deficits were most pronounced early in the block and diminished over time, consistent with a gradual resolution of goal competition.

Critically, a significant interaction between reward and switching was observed, $\beta = 0.260$, $SE = 0.098$, $p = .008$, indicating that the detrimental effect of switching on accuracy was reduced under rewarded conditions. This effect was further qualified by a significant three-way interaction among trial number, reward, and switching, $\beta = 0.042$, $SE = 0.017$, $p = .012$. The negative direction of this interaction indicates that the beneficial effect of reward on switch trials was strongest early in the block and diminished over time, suggesting that reward particularly improves performance when attentional demands and goal competition are highest (Figure 4).

4 Discussion

4.1 N2pc results

We predicted that N2pc amplitudes would be larger (more negative) on no-go trials and further modulated by task switching and reward, reflecting increased attentional selection under conditions of increased goal competition. A $2 \times 2 \times 2$ repeated-measures ANOVA on N2pc amplitude revealed a significant main effect of Trial Type, $F(1, 28) = 38.47$, $p < .001$, indicating that N2pc amplitudes differed between go and no-go trials. Follow-up comparisons confirmed that N2pc was significantly more negative for no-go trials than for go trials across both task conditions (repeat and switch; all $ps < .001$) and across reward levels (rewarded and unrewarded; all $ps < .001$), reflecting stronger lateralized attentional selection during response inhibition demands.

There was no significant main effect of Condition, $F(1, 28) = 2.96$, $p = .097$, nor a main

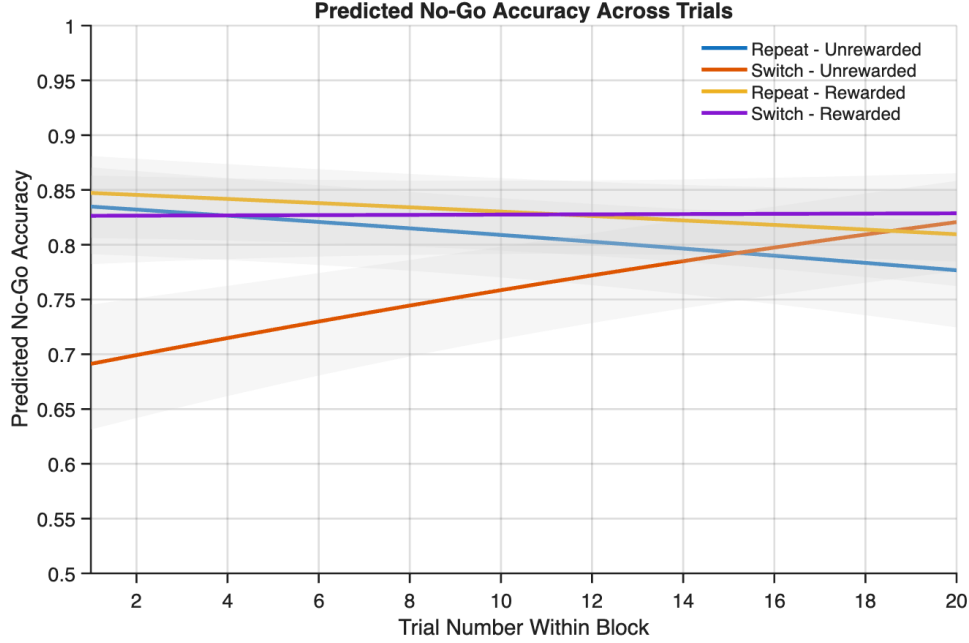


Figure 4: Predicted No-Go accuracy across trials as a function of task switching and reward.

effect of Reward, $F(1, 28) = 0.09$, $p = .764$. Importantly, none of the interactions reached statistical significance, including Trial Type $\times$ Condition, $F(1, 28) = 2.29$, $p = .142$, Trial Type $\times$ Reward, $F(1, 28) = 0.24$, $p = .631$, Condition $\times$ Reward, $F(1, 28) = 0.54$, $p = .468$, and the three-way interaction, $F(1, 28) = 1.63$, $p = .212$.

4.2 Contralateral and Ipsilateral Activity

Because the N2pc reflects the difference between contralateral and ipsilateral activity, it provides a measure of lateralized attentional selection but may mask effects that are present in both hemispheres. In particular, if task manipulations influence both target-related (contralateral) and distractor-related (ipsilateral) processing in similar directions, these effects may cancel out in the N2pc. To address this possibility and better characterize the underlying neural mechanisms, separate $2 \times 2 \times 2$ repeated-measures ANOVAs were conducted on contralateral and ipsilateral activity. For contralateral activity, there were significant main effects of Trial Type, $F(1, 28) = 8.62$, $p = .007$, Condition, $F(1, 28) = 9.00$, $p = .006$, and Reward, $F(1, 28) = 11.08$, $p = .002$. These results indicate that neural responses at electrodes contralateral to the target were modulated by task demands, with differential processing between go and no-go trials, as well as between switch versus repeat trials and rewarded versus unrewarded conditions. In addition, significant Trial Type $\times$ Condition, $F(1, 28) = 11.48$, $p = .002$, and Trial Type $\times$ Reward interactions, $F(1, 28) = 16.02$, $p = .001$.

.001, suggest that the influence of switching and reward on target-related processing differed between go and no-go trials. A similar pattern was observed for ipsilateral activity. There were significant main effects of Trial Type, $F(1, 28) = 36.41$, $p < .001$, Condition, $F(1, 28) = 6.54$, $p = .016$, and Reward, $F(1, 28) = 6.71$, $p = .015$, indicating that activity on the distractor side was also sensitive to task demands and motivational context. Significant Trial Type $\times$ Condition, $F(1, 28) = 10.00$, $p = .004$, and Trial Type $\times$ Reward interactions, $F(1, 28) = 6.46$, $p = .017$, further indicate that these effects varied as a function of behavioral context. (Figure 5) (Figure 6) Despite these robust effects in both contralateral and ipsilateral signals, the N2pc did not show significant modulation by condition or reward. This pattern can be explained by the fact that both contralateral and ipsilateral activity were modulated in similar directions across conditions. Specifically, task demands and reward influenced not only in contralateral activity, but also in ipsilateral activity. Because the N2pc reflects the difference between these two signals, parallel changes in contralateral and ipsilateral amplitudes effectively reduced the net difference, leading to an absence of significant effects in the N2pc measure. These findings suggest that reward and task switching modulate early attentional processing in a non-lateralized manner, influencing both target enhancement and distractor processing simultaneously, rather than selectively biasing spatial attention as indexed by the N2pc.

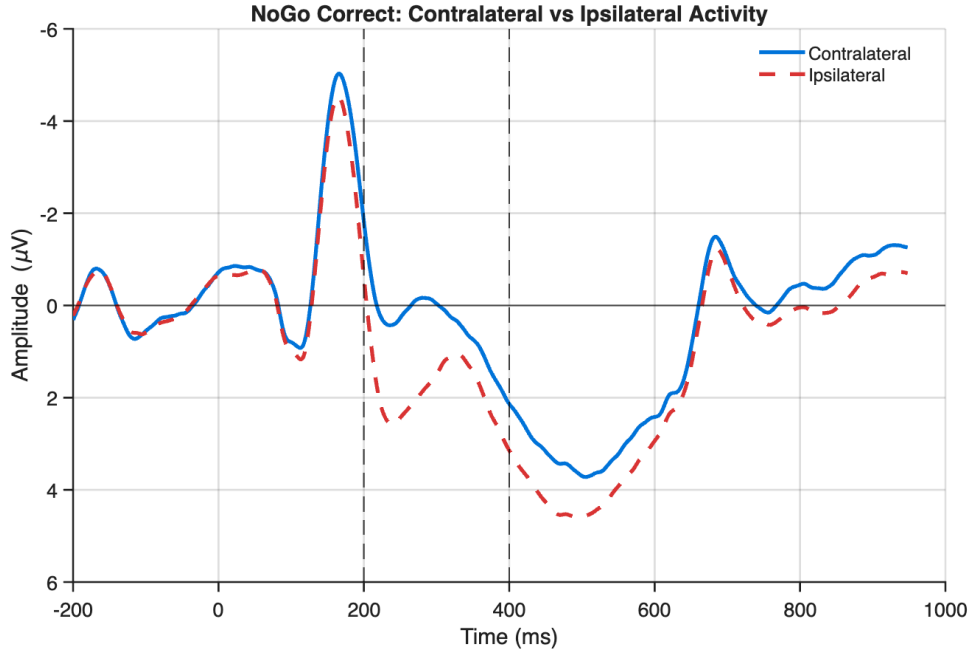


Figure 5: Contralateral and ipsilateral activity for no-go correct trials.

5 Discussion

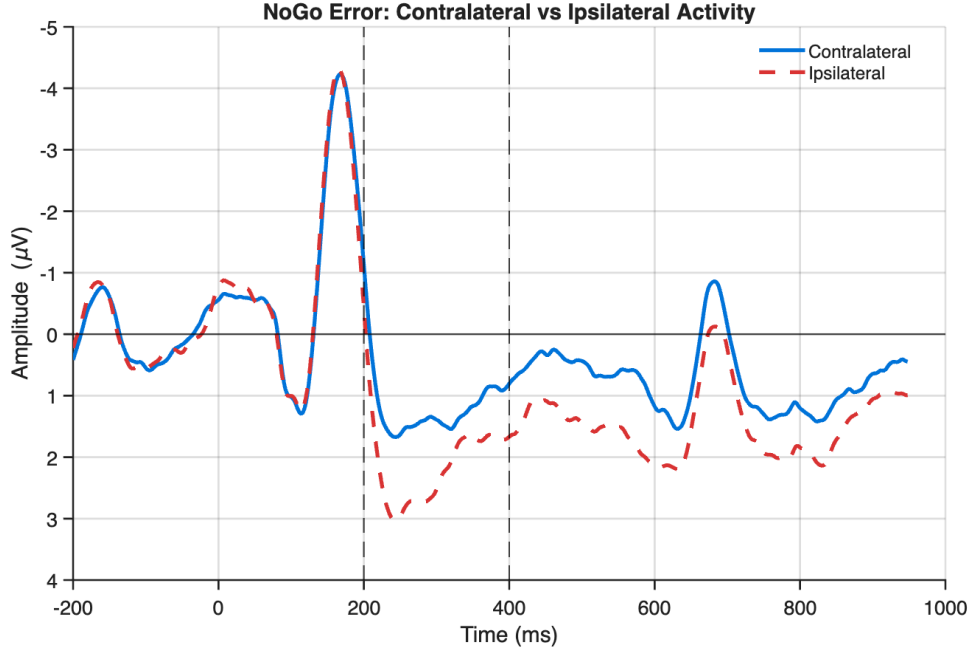


Figure 6: Contralateral and ipsilateral activity for no-go error trials.

5.1 P3 results

Based on the hypothesis that goal competition and motivational context influence attentional engagement, we predicted larger P3 amplitudes for no-go trials, with modulation by task switching and reward. A $2 \times 2 \times 2$ repeated-measures ANOVA was conducted on P3 amplitude with factors of Trial Type (go vs. no-go), Condition (repeat vs. switch), and Reward (rewarded vs. unrewarded). The analysis revealed significant main effects of Trial Type, $F(1, 28) = 37.57$, $p < .001$, Condition, $F(1, 28) = 48.75$, $p < .001$, and Reward, $F(1, 28) = 30.60$, $p < .001$. P3 amplitudes were larger for no-go than go trials, larger in repeat than switch conditions, and larger for rewarded than unrewarded trials, indicating that attentional engagement was enhanced during response inhibition, reduced under task-switching demands, and increased under motivational incentives.

There were significant interactions between Trial Type and Condition, $F(1, 28) = 34.40$, $p < .001$, and between Trial Type and Reward, $F(1, 28) = 22.84$, $p < .001$. Follow-up comparisons were conducted to further characterize these interactions. For the Trial Type $\times$ Condition interaction, P3 amplitude was significantly reduced on switch relative to repeat trials for both trial types; however, this reduction was substantially larger for no-go trials

(mean difference = 1.32, $p < .001$) than for go trials (mean difference = 0.24, $p = .012$). For the Trial Type $\times$ Reward interaction, reward significantly increased P3 amplitude for both trial types, with a markedly larger effect for no-go trials (mean difference = 2.16, $p < .001$) than for go trials (mean difference = 0.61, $p = .005$).

Additionally, P3 amplitude was significantly larger for no-go compared to go trials across both conditions (repeat and switch; all p s $< .001$) and reward levels (rewarded and unrewarded; all p s $< .001$), confirming a robust effect of Trial Type. The Condition $\times$ Reward interaction was not significant, $F(1, 28) = 0.001$, $p = .972$, nor was the three-way interaction among Trial Type, Condition, and Reward, $F(1, 28) = 0.10$, $p = .751$.

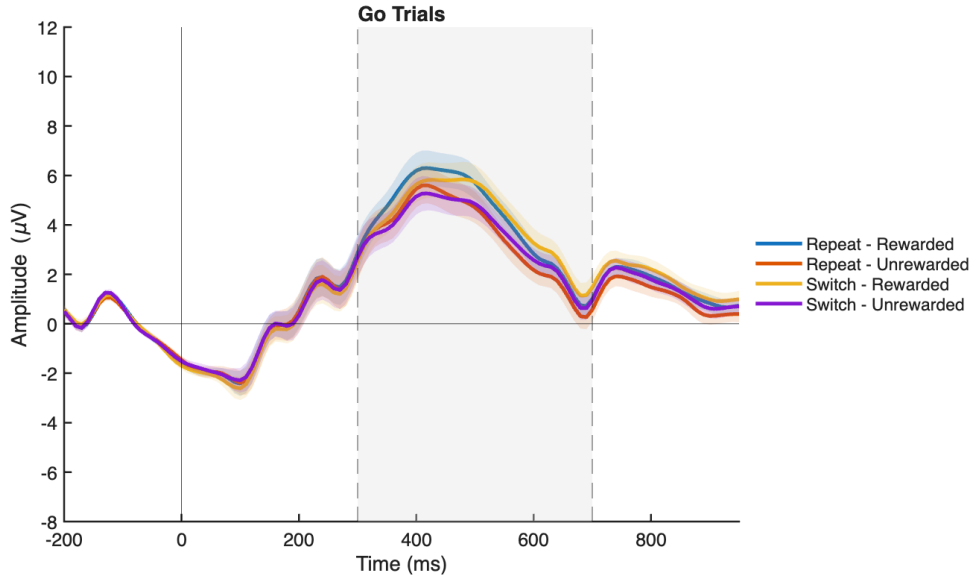


Figure 7: P3 waveforms at Pz by reward and task-switching condition. Grand-average ERP waveform for go trials

5.2 Oscillatory Power Results

To examine oscillatory dynamics associated with attentional lapses, linear mixed-effects models were conducted on subject-level estimates of alpha and theta power derived from no-go trials.

Theta-band activity (4–7 Hz) Analysis of frontocentral theta power during the post-stimulus window (0–950 ms) revealed a significant main effect of accuracy, $F(1, 224) = 36.91$, $p < .001$, such that no-go error trials elicited greater theta power than correct trials. There were no significant main effects of condition, $F(1, 224) = 3.57$, $p = .060$, or reward,

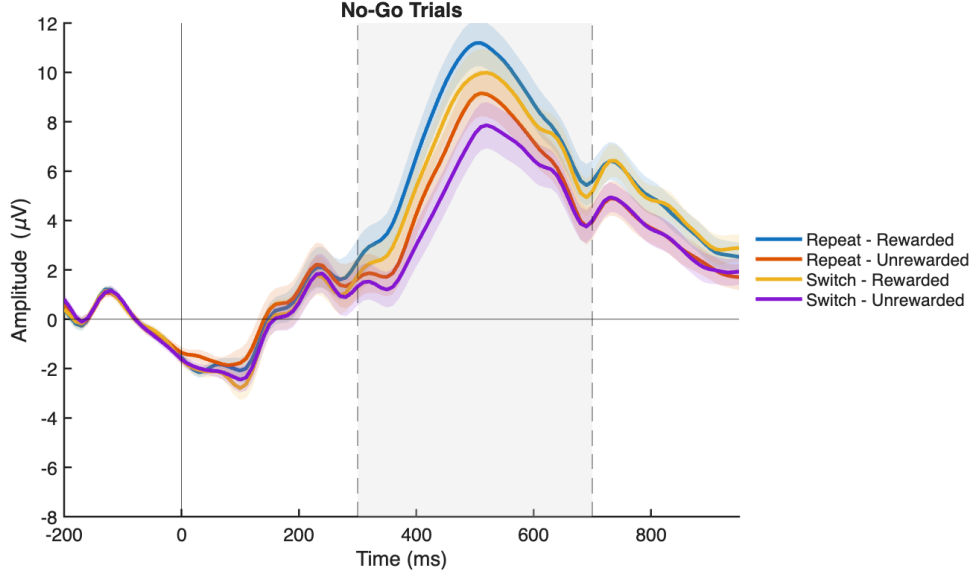


Figure 8: P3 waveforms at Pz by reward and task-switching condition. Grand-average ERP waveform for no-go trials

$F(1, 224) = 3.25$, $p = .073$, although both effects trended toward significance. No significant interactions were observed (all p s $> .22$).

Alpha-band activity (8–14 Hz) Pre-stimulus alpha (200 to 0 ms) did not differ as a function of accuracy, $F(1, 224) = 0.27$, $p = .60$, condition, $F(1, 224) = 0.001$, $p = .97$, or reward, $F(1, 224) = 0.70$, $p = .40$, and no interactions were significant (all p s $> .50$). Similarly, post-stimulus alpha (0–950 ms) showed no significant main effects of accuracy, $F(1, 224) = 0.43$, $p = .51$, condition, $F(1, 224) = 0.54$, $p = .46$, or reward, $F(1, 224) = 0.92$, $p = .34$, and no significant interactions (all p s $> .43$).

Together, these results indicate that frontocentral theta power reliably differentiates no-go errors from correct responses, whereas posterior alpha power does not show sensitivity to attentional lapses or task manipulations.

5.3 MVPA Decoding

A multivariate pattern analysis (MVPA) was conducted to examine whether reward condition (rewarded vs. unrewarded) could be decoded from trial-level EEG activity. For each trial, the classifier was trained on a multivariate feature vector consisting of mean EEG amplitude within the 0–500 ms post-stimulus window at each electrode, thereby capturing the spatial distribution of neural activity across the scalp.

Using a within-subject cross-validation approach (80/20 train–test split, repeated 20

times), classification accuracy was significantly above chance ($M = 0.549$, $SEM = 0.005$), $t(28) = 9.59$, $p < .001$. As shown in Figure, decoding performance was consistently above chance across participants, with most accuracies ranging between 0.52 and 0.60, indicating that reward-related information could be reliably decoded from individual participants' neural activity patterns.

To assess the extent to which reward-related representations generalized across participants, a complementary leave-one-subject-out (LOSO) analysis was conducted. In this analysis, the classifier was trained on the spatial pattern of EEG activity from all but one participant and tested on the held-out participant. Decoding accuracy remained significantly above chance ($M = 0.517$, $SEM = 0.003$), $t(28) = 5.64$, $p < .001$, although the magnitude of the effect was reduced relative to the within-subject analysis.

Together, these findings indicate that reward-related information is encoded in distributed patterns of EEG activity. The stronger decoding performance observed within individuals suggests that these representations are robust at the individual level, while the reduced cross-subject accuracy reflects inter-individual variability in the spatial organization of neural activity. The modest effect size observed in the cross-subject analysis is consistent with prior EEG decoding studies and reflects the low signal-to-noise ratio of scalp-recorded data and the distributed nature of cognitive representations.

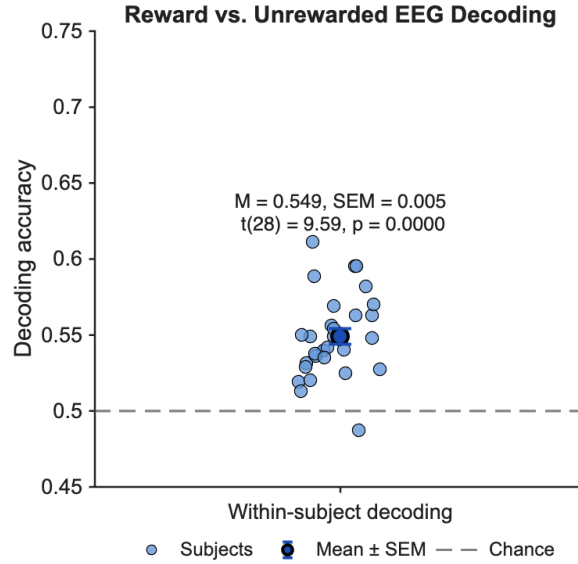


Figure 9: rewarded vs. unrewarded within-subject decoding

6 Discussion

6.1 Overview of Findings

The present study investigated how reward and task switching shape sustained attention, with a particular focus on how goal competition contributes to attentional lapses. Across behavioral and neural measures, the findings converge on a clear pattern: attentional failures are not random breakdowns, but emerge systematically during periods of heightened competition between task goals, and can be selectively mitigated by reward.

Behaviorally, switching between task goals increased no-go errors, particularly when switching to an unrewarded goal. Trial-level analyses revealed that these switch costs were strongest early in the block and diminished over time, with reward exerting its strongest benefits during this early period. Neural results further revealed a dissociation across processing stages. Early attentional selection (N2pc) showed limited sensitivity to reward and switching at the lateralized level, whereas later control-related processing (P3) was strongly modulated. Oscillatory power analyses indicated that frontocentral theta reliably differentiated errors at the moment of response. Together, these findings support a dynamic account of sustained attention in which reward improves performance particularly during periods of high competition, while attentional failures are most clearly expressed at later stages of processing associated with response selection and control, rather than early perceptual spatial attention.

Importantly, these results are consistent with recent EEG work suggesting that attentional lapses reflect changes in overall neural coordination rather than simple failures of selective attention. For example, Chidharom, Jones, et al. (2025) showed that distraction is associated with reduced large-scale neural coordination and attenuated ERP responses, even when participants are visually engaged. In this context, the present findings suggest that lapses are more closely related to variability in task engagement and control processes than to failures of early attentional selection.

6.2 Behavioral Evidence of Goal Competition

The behavioral results provide strong support for the goal-competition framework. Participants exhibited higher no-go error rates on switch trials than repeat trials. This finding reflects a classic switch cost, where following a switch, the prior task set remains partially active, creating a period of instability during which attention is more vulnerable to lapses (Chidharom, Vogel, & Rosenberg, 2025). Within this framework, a switch induces a period of instability where the prior task set remains partially active, creating proactive interference. This competition between the previous and current goals renders attention significantly more vulnerable to lapses, or “out-of-the-zone” states (Esterman et al., 2014).

Importantly, this switch cost was modulated by reward context: switching increased errors only when the current goal was unrewarded. In contrast, switching toward a rewarded goal largely eliminated this cost, suggesting that motivational incentives facilitate “goal shielding.” In these instances, the prospective reward may act as a buffer that prioritizes the current task set and suppresses competing representations from the previous block (Chidharom, Vogel, & Rosenberg, 2025).

Trial-level analyses revealed that these effects are time-dependent within a block. Switch costs were largest at the beginning of a block and diminished over time, suggesting that the behavioral consequences of switching are transient and recover as participants gradually adapt to the current task demands. This recovery likely reflects the gradual strengthening of the new goal’s representation, allowing participants to transition from reactive error-correction to a stable, proactive attentional state (Chidharom, Vogel, & Rosenberg, 2025). Furthermore, reward selectively reduced errors during this early period, indicating that motivational incentives are most effective when attentional demands are highest.

6.3 Early Attention: Limited Role of Spatial Selection

The N2pc results suggest that early spatial attention was modulated by reward and switching effects. While N2pc amplitude was larger for no-go than go trials, reflecting increased attentional selection under inhibitory demands, it was not significantly modulated by condition or reward. This indicates that the behavioral effects are unlikely to be driven by selective allocation of spatial attention to the relevant stimulus.

However, follow-up analyses of contralateral and ipsilateral activity revealed that both signals were significantly modulated by task condition and reward. This suggests that early visual processing was influenced by these factors, but in a non-lateralized manner. Because the N2pc reflects the difference between contralateral and ipsilateral activity, parallel changes in both signals may have canceled out in the difference wave.

These findings are consistent with recent work suggesting that attentional lapses are not fully explained by failures of early selective attention. For example, Chidharom, Jones, et al. (2025) showed that distraction-related neural signatures can be dissociated from classic markers of spatial attention, indicating that participants may process stimuli perceptually while failing to fully engage with task demands. In this context, the present results suggest that the effects of task switching and reward are not primarily reflected in spatial biasing of attention, but in broader changes in processing.

6.4 Late Control Processes: Reward Stabilizes Goal Engagement

In contrast to early attentional processing, the P3 results provide strong evidence that reward and switching modulate later-stage control processes. P3 amplitude was larger for no-go than go trials, reduced under switching, and enhanced by reward, with particularly strong effects for no-go trials. Importantly, switching-related reductions in P3 were larger for no-go trials than for go trials, indicating that switching is associated with reduced engagement under conditions of high goal competition. Similarly, reward-related increases in P3 were more pronounced for no-go trials, suggesting that motivational incentives preferentially enhance engagement under higher control demands.

These findings are consistent with theories linking the P3 to context updating and the allocation of cognitive resources (Donchin & Coles, 1988; Nieuwenhuis et al., 2005). Rather than reflecting changes in early perceptual selection, the present results suggest that reward is associated with increased task engagement during response selection, particularly when demands on inhibitory control are high.

6.5 Oscillatory Dynamics

The oscillatory results further clarify the timing of attentional lapses. Frontocentral theta power was significantly greater for no-go errors than correct trials, consistent with prior work linking theta activity to conflict monitoring and control-related processing (O’Connell et al., 2009). This pattern is consistent with the idea that theta reflects reactive processes engaged when control demands are not successfully resolved.

In contrast, no alpha showed sensitivity to accuracy, condition, or reward. Although prior studies have linked alpha-band activity to fluctuations in attentional engagement (O’Connell et al., 2009), such effects were not observed here. This discrepancy may reflect differences in task structure, as the relatively short pre-stimulus window and rapid trial sequence may limit sensitivity to slower fluctuations in attentional state.

6.6 Neural Representation of Reward Context

The MVPA results demonstrate that reward-related information is encoded in distributed EEG activity and can be reliably decoded above chance, both within and across participants. This indicates that reward conditions are associated with differences in large-scale patterns of neural activity.

This finding is consistent with prior work showing that attentional states can be characterized by distributed neural patterns. For example, Chidharom, Vogel, and Rosenberg (2025) demonstrated that EEG signals can distinguish between different attentional states,

while work using naturalistic paradigms has shown that attentional engagement is associated with increased neural synchrony across individuals (Song et al., 2021).

6.7 Implications and Limitations

Taken together, the findings support a dynamic view of sustained attention in which performance varies as a function of task demands and motivational context. In particular, switching between task goals is associated with transient increases in attentional lapses, and reward is associated with improved performance during periods of increased goal competition. At the neural level, these effects are most strongly reflected in later-stage processing associated with response selection and control, rather than early spatial and perceptual selection. Early visual processing appears broadly sensitive to task demands and reward, but not in a spatially selective manner. Oscillatory analyses further indicate that lapses are most clearly reflected at the time of stimulus processing rather than being reliably predicted in advance.

Several limitations should be considered. First, the relatively short pre-stimulus window may have limited sensitivity to slower fluctuations in attentional state. Future studies could incorporate longer intertrial intervals or continuous paradigms to better capture proactive attentional dynamics. Second, the block-based reward manipulation may confound sustained motivational state with trial-level dynamics. Third, although decoding revealed reliable reward-related differences in neural activity, effect sizes were modest, reflecting the inherent limitations of EEG.

Future work could combine time-resolved neural decoding with trial-level behavioral modeling to better characterize how neural and behavioral dynamics interact over time, and how motivational factors influence these processes under varying task demands.

Data and Code Availability Statement

[Here](#) is the GitHub repo used in our analysis.

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